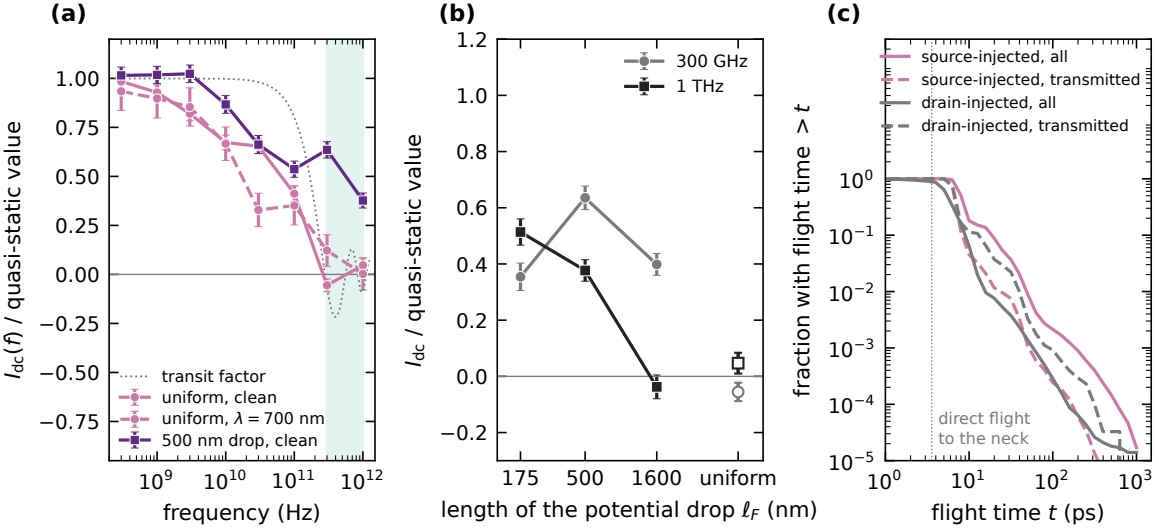
(a)
$I_{dc}(f)$ / quasi-static value
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
transit factor
uniform, clean
uniform, $\lambda = 700$ nm
500 nm drop, clean
10^9
10^{10}
10^{11}
10^{12}
frequency (Hz)
(b)
I_{dc} / quasi-static value
1.2
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
300 GHz
1 THz
175
500
1600
uniform
length of the potential drop ℓ_F (nm)
(c)
fraction with flight time $> t$
source-injected, all
source-injected, transmitted
drain-injected, all
drain-injected, transmitted
10^0
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
direct flight
to the neck
10^0
10^1
10^2
10^3
flight time t (ps)